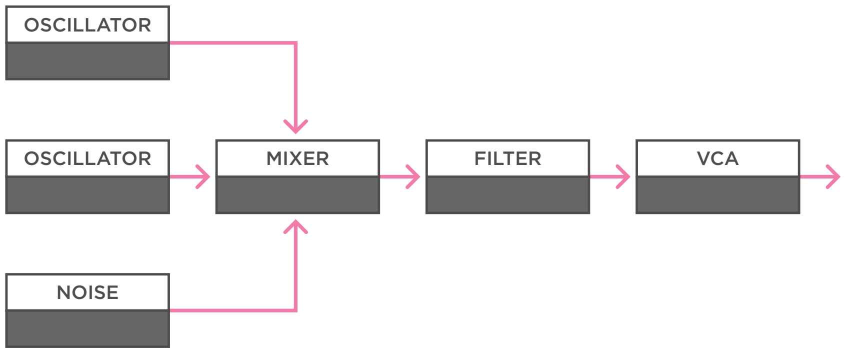
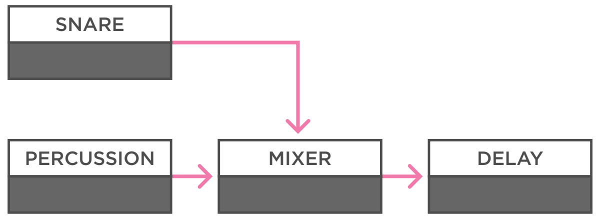
Modular Utilities

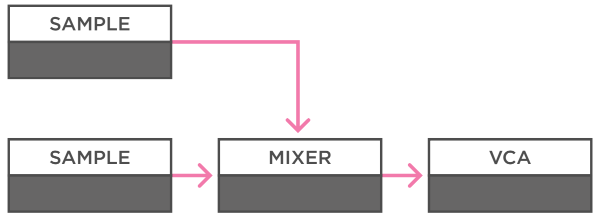
Audio Mixing

Audio
Control Voltage
Trigger / Gate





(b)



(a)

(c)

a - The most common function for a mixer in a synth-voice is to mix different oscillators and noise sources together before they go to a filter for further sound shaping.

b - Besides using a mixer in a main synth-voice you can use them to combine other audio signals within a larger patch. For example, mix two percussive elements before they go to an effects module.

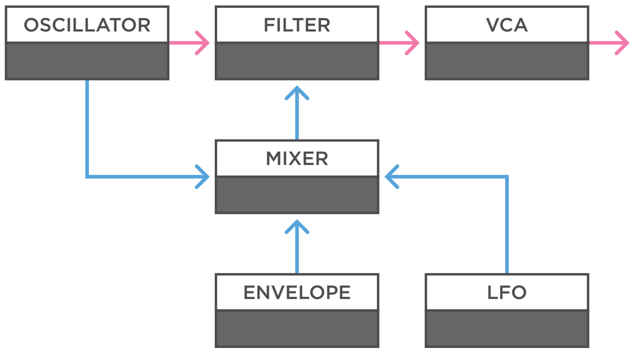
c - Or mix two background samples like field recordings with different lengths, before they go to a VCA to shape the volume.

Modular Utilities

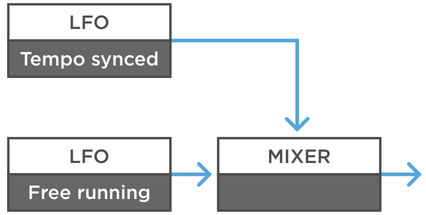
CV Mixing

Audio
Control Voltage
Trigger / Gate

MONOTRAIL
TECH TALK



(a)



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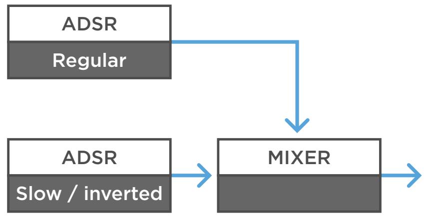

(c)

a - Instead of using a mixer to combine audio signals, you can use them to combine control voltages. Take a filter for example. A common thing to do is feed it an envelope, but you can make it more interesting by mixing an envelope with a slow LFO for some dynamics over time. I even like adding a third modulation by mixing-in a separate waveform output of the main oscillator. This can really add nice textures to your sound.

b - Mixing two modulation sources together can create lovely results as well. Take a tempo synced and a slow free running LFO for example.

c - Or try creating new envelope shapes by mixing two ADSR envelopes with different settings and polarity together.

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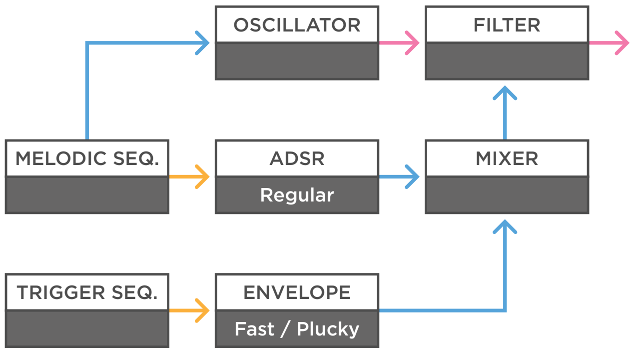
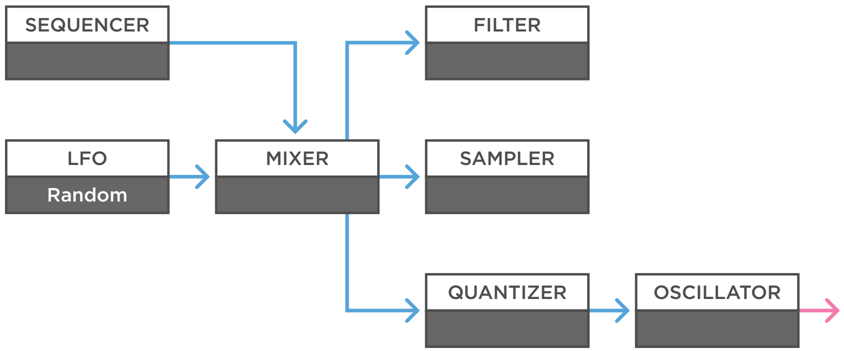
MONOTRAIL TECH TALK



(a)



(b)

a - You can mix a regular envelope to the filter of your synth voice, with a quick one, triggered by a sequencer for example, to add clicky percussive elements to your basic synth patch.

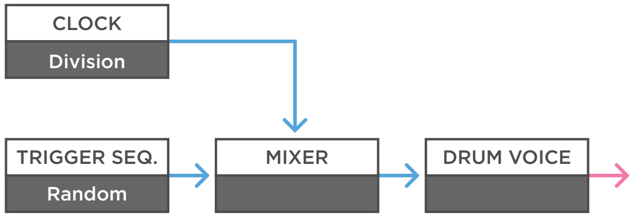
b - Combining the output of a sequencer with a subtle signal from an LFO or smooth random voltage can also lead to great results. Send them to something like a filter, a sample player, or even a quantizer, in order to generate melodies.

Modular Utilities

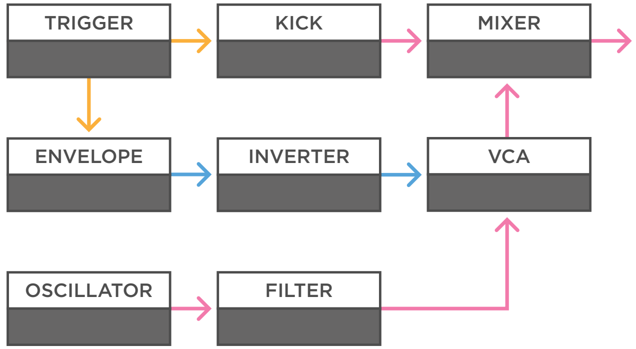
Mixing / Inverting

Audio
Control Voltage
Trigger / Gate

MONOTRAIL
TECH TALK



(a)



(b)

a - You can also use a mixer as a trigger combiner. For example, to combine a steady trigger from a clock divider or sequencer, with a sporadic random trigger, before sending it to a drum module.

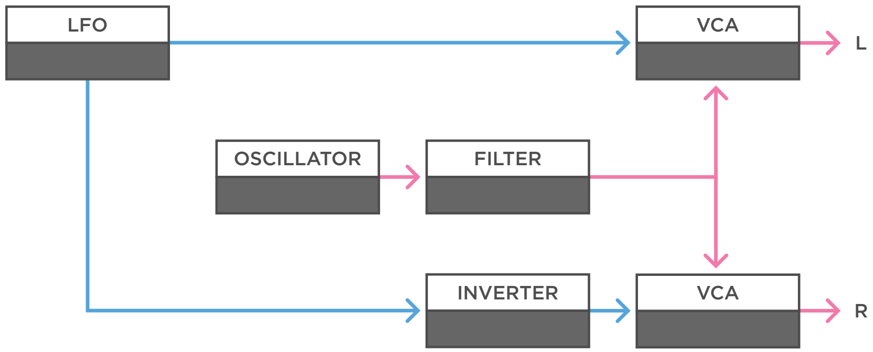
b - Inverting signals can be powerful. For example, to create a sidechain effect. In this example we have a sequencer triggering a kick drum. That kick is mixed with a baseline created with a simple oscillator, filter, and VCA setup. When we have the sequencer trigger a short attack decay envelope as well, and we invert that signal before going to the VCA of the baseline, the volume of the baseline gets pushed down by the envelope, every time the kick hits.

Modular Utilities

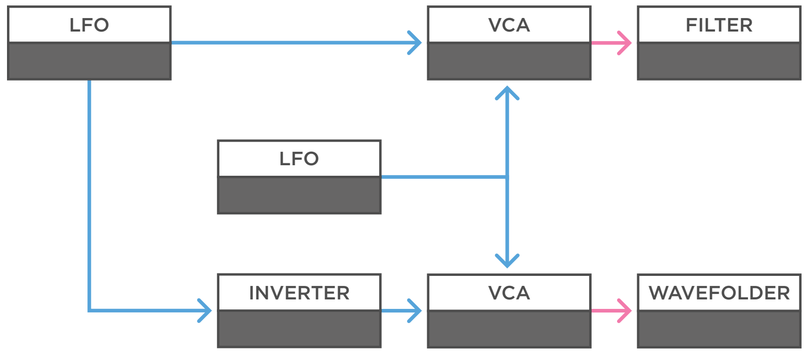
Inverting

- Audio
- Control Voltage
- Trigger / Gate

MONOTRAIL
TECH TALK



(a)



(b)

a - Another trick is to create stereo panning by multiplying a signal like a synth-voice, to two half-opened VCAs. To create some motion in the stereo field we multiply a modulation source, like an envelope or LFO, and send the original to one of the VCAs, and an inverted one to the other. This way if one signal opens, the other closes, and the other way around.

b - This trick is also a great way to have something like an LFO crossfade one modulation source between two destinations.